information or challenge the decision.

5. Finding Errors and Bias.

6. Legal and professional Responsibility.

🕒 ; 7 min

(b) Discuss at least two societal harms that may occur when an AI system operates as black box.

A black box AI system is a system where people cannot easily understand how the AI makes its decision. They can create several harms in society.

1. Discrimination and Bias.

2. Example: In the given loan system, if applicants from a particular community are rejected

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more often without a clear reason:

2. Lack of Accountability: When a AI decision cannot be explained, it becomes difficult to determine who is responsible when something goes wrong.

3. Loss of trust: People may lose trust in organization when important decision are made by system.

4. Difficulty in Correcting Erros:

: 7 min

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(c) Explain how lack of explainability can effect procedural fairness, accountability, public trust and an individual's ability to challenge

Lack of explainability means people cannot clearly understand why an AI system made a particular decision.

1. Procedural fairness: people should be treated fairly during the decision making process.

2. Accountability: when the AI's decision cannot be explained.

3. public trust: people are less likely to trust an AI system when they donot understand how it makes important decisions.

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4. Ability to Challenge a Decision: A person needs to know the reason for a decision in order to correct errors or challenge it.

⏱: 6 min

3(d) Discuss what information should ideally be provided to an affected person when an AI system makes an important decision about them.

When an AI system makes an important decision about a person. the affected person should receive clear and understandable information about the decision.

1. Reason for the Decision: The person should know why the AI Made the decision.

2. Important Factors used: The organization should explain the main factors or type of

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data that influenced the decision.

3. Use of AI: The person should be informed that an AI or automated system was involved in making the decision.

4. possible error: They should be told how incorrect or outdate information could affect the decision.

5. Right to Review or Appeal.

6. How to correct information.

7. Contact Information.

: 7min

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(4) ACM code of Ethics and Professional Decision Making

(a) Explain the key principle of the ACM code of Ethics and why professional codes are important for computing networki professinals.

The ACM (Association for Computing Machinery) Code of ethics is a set of professional guidelines.

key principle of the ACM code of Ethics

1. Contribute to society and Human well-Being.

2. Avoid Harm: professionals should prevent actions that may harm user, organizations.

3. Be Honest and Trustworthy.

4. Respect privacy : personal information should be collected.

5. Maintain security and confidentiality.

6. personal competence :

Importance of professional codes :

- provide guidance for making ethical decision.
- Helps professional act responsibily and consistently.
- protect users, organizations and society form harm.
- promote honesty, fairness, privacy and security.

: 7 min

(b) Identify the ACM principles that are relevant to the above situation and explain how they should influence the engineer's decision.

In this situation the manager wants to release the system even through testing has found serious security weakness.

1. Contribute Society and Human Well-Being.

2. Avoid Harm.

3. Be Honest and Trustworthy.

4. Respect privacy

5. Maintain professional Competence

6. ..

: 5min

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(c) Discuss the responsibilities of software professionals when organization pressure conflicts with public safety. Security or user interests.

Software professionals have important responsibilities when organization pressure conflicts with public safety, security or user interest.

1. protect public safety.

2. protect Security.

3. protect user interests.

4. Be honest.

5. Raise ethical concerns.

6. use professional judgment.

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7. Refuse Harmful Action when necessary.

: 7min

(d) Explain how professional code can support ethical decision making without replacing an individual professional judgment.

⇒ How a professional code support Decision Making.

1. provides Guidance.

2. Helps Identify Ethical problem.

3. provides a common Standard.

4. Support Difficult Decision.

5. Encourage Responsibility.

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Why professional judgment is still

Aa Necessary.

A code can not provide an exact answer for every situation. Different situation may involve different risks, laws, users and ethical conditions. Therefore, engineers must use their own knowledge experience, critical thinking and understanding of the situation.

: 7min

5. Whistleblowing , Professional Dissent and Ethica Responsibility.

(a) Define professional dissent and whistleblowing in the context of computing.

professional Dissent :

professional Dissent means respectfully disagreeing with a decision or practice at the workplace because it may be unsafe, unethical or harmful. A Software engineer may rise concerns with a manager or team when they believe a decision could harm user.

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Whistleblowing:

Whistleblowing means reporting serious wrongdoing illegal activities or dangerous practice to an appropriate authority, especially when normal internal channels do not solve the problem.

: 7min

(b) Explain the ethical considerations that the analyst should evaluate before deciding wether to disclose the issue externally.

Before deciding to disclose the issue externally, the cybersecurity analyst should carefully evaluate several ethical factors:

1. Severity of Harm: The analyst should determine how serious the vulnerability is and wether it could cause significant harm to customers or the public.

2. Public saftey and Security.

3. Internal Reporting effort.

4. Evidance and Accuracy.

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5. Legal and professional obligations.

6. Possible Consequence : The analyst should use ap the effects of disclosure on customers the company.

7. Appropriate Disclosure Method: If external reporting because necessary the Analyst should use appropriate and lawful channels to minimize unnecessary harm.

: 7 min

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6(a) Explain how the ACM principles "Contribute to society and human well-being" and Respect privacy " may come into conflict.

The ACM code of Ethics says computing professionals should contribute to society and human well-being and also repect privacy.

1. Contribute to society and human well-being.

2. Respect privacy.

Why they conflick:

The research may provide a large social benefit, but collecting or analyzing detailed location

Example: If researchers use location data to identify areas where an infectious disease is spreading, the information could help authorities respond quickly.

Ethical Balance:

- use anonymized or aggregated data.
- Collect only the data that is necessary
- Limit access to sensitive information.
- use strong security measures.
- Clearly define how the data will be used and stored.

: 8 min

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(b) Discuss the ethical principles that should guide the collection and use of personal data, including information consent, purpose limitation, data minimization, security transparency and accountability.

When collecting and using personal data, data scientists should follow important ethical principle to protect individuals and prevent misuse of their information.

1. Informed consent.

2. purpose Limitation.

3. Data Minimization.

4. Security

5. Transparency.

6. Accountability.

: 5min

7. Copyright, Patents, and Trademarks in Computing.

(a) Differentiate among copyright, patents and trademark in the context of software and digital products.

Copyright, Patents and trademark are different forms of intellectual property protection. They protect different parts of a software or digital product.

Copyright	Patent	Trademark
original creative and expression	A new and useful invention or technical method	A name, logo, symbol, or brand that indentifies a product
Source code, documentation, website design	A new hardware-assisted algorithm or technical process	A software company's product name and logo

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Example: If a startup writes original source code for it Software platform, another cannot simply copy that code and present it as it own.	Example: The Startup's new hardware-assisted algorithm may potentially qualify for patent Protection if it meets the required legal condition.	Example: The startup can use trademark protection for its distinctive product name and logo.

: 8 min

(b) Identify which form of intellectual-property protection is most appropriate for the source code, the novel technology invention, and the product name / logo; justify each choice.

For the startup's software platform, different parts should be protected using different forms of intellectual property.

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1. Source code → copyright

Copyright is the most appropriate protection for the original Source code because it protect the expression of an original Software program.

2. Novel Technology Invention → patent

A patent may be appropriate for the new hardware-assisted algorithm if it meets the applicable legal requirments for patentability.

3. Product Name and Logo → Trademark

Trademark is appropriate for the distinctive product name and logo because theyindentify the Startup's products or services and distinguish them from Competitors.

: 8min

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8: Open-Source and proprietary Software Licensing.

(a) Compare and Contrast open-source software licensing and proprietary software licensing.

Open-source software	proprientary software
(i) usually available for users to inspect.	(i) usually kept private.
(ii) users can modify it according to the license	(ii) Modification is usually restricted.
(iii) Redistribution is generally allowed under licens conditions.	(iii) usally controlled by the owner.
(iv) Often free to use, but not always.	(iv) often requires payment or subcription
(v) Most community involvement.	(v) Mainly controlled by the company.

🕒 : 7 min

(b) Discuss the ethical advantages and potential ethical concerns of each licensing model.

1. Open-Source Software:

Ethical advantages:

- Transparency: Users and researchers can inspect the source code.
- Collaboration: Developers can work together to improve the software.
- Knowledge Sharing: Students and researchers can learn from the code.
- Security Review: More people can inspect the code and identify security problems.

Accessibility: Software can often be used by more people under the license terms.

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2. proprietary Software:

Ethical Advantages:

• Control: The owner can control how the software is distributed and used.

• Quality Control: The owner can control official versions and updates.

Ethical Concerns:

• Limited Transparency: Users cannot easily inspect how the software works.

• Vendor Dependence: Users may become dependent on one company for updates and support.

: 8 min

🕒: 7min IT23039

(9) Responsible Disclosure of a cybersecurity Vulnerability.

(a) Define responsible disclosure and distinguish it form unauthorized exploitation or cybercrime.

Responsible Disclosure	Unauthorized Exploitation
(i) Reports the vulnerability to the organization.	(i) Uses the vulnerability without permission.
(ii) Avoids accessing real user's private data.	(ii) May access, steal, or misuse private data.
(iii) Uses only the minimum testing needed to confirm the issue.	(iii) May cause damage or disruption
(iv) Gives the organization time to fix the problem.	(iv) May exploit or publicly expose the vulnerability.
(v) Main goals is to protect users and improve security	(v) May involve harm, theft or unauthorized access.

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(b) Explain the ethical guidelines that the researcher should follow after discovering the vulnerability.

After discovering a serious vulnerability, ethical guidelines:

1. Do Not Access Private Data:

2. Minimize Testing.

3. Reports to the Organization.

4. provide Clear Evidence.

5. Maintain Confidentiality.

6. Give Reasonable Time to fix.

7. protect users.

8. Follow Legal and professional Rules.

: 7 min

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Ⓛ:7min IT 23039

10. Ethical Hacking Versus Cybercrime—

(a) Differentiate between ethical hacking and cybercrime.

Ethical Hacking	Cybercrime
(i) Done with proper authorization.	(i) Done without permission
(ii) Goal is to find and fix Security weakness.	(ii) Goal be to steal, damage disrupt or misuse data.
(iii) Follows the agreed scope and rules.	(iii) Ignores authorization and legal boundaries.
(iv) Protects users and organizations.	(iv) Can cause harm to users or organization.
(v) Findings are normally reported to the organization.	(v) Stolen or accessed information may be misued or exposed.

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(b) Explain the conditions under which hacking activities may be considered ethical, including, authorization, scope, purpose, proportionality and respect for privacy.

1. Authorization: The organization must give clear permission to perform the security testing.

2. Scope: The tester must test only the system applications and ideas.

3. Purpose: The main purpose should be to identify and fix security weaknesses.

4. proportionality: Testing should be limited.

5. Respect for privacy: Tester should avoid accessing personal.

Time: 7min

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(11) Algorithmic Bias in Recruitment :

(a) Define Algorithm bias and explain at least four possible causes, such as biased training data, historical dicrimination.

Causes of Algorithmic Bias :

1. Biased Traning Data.

2. Historical Discriminiation.

3. Proxy variable.

4. sampling problem

5. Inappropriate Model objeetives.

The system may be designed to maximize a goal such as predicting which previous application were hired.

: 7min

(b) Explain how bias can be dected and measured during AI development and deployment.

Bias can be detected during both AI development and deployment.

1. Compare selection Rates: check wether different demographic groups are selected or rejected at significantly.

2. Compare Accuracy: Measure whether the AI make carrect predictions equally.

3. Check False positive and false Negative Rate.

4. Test the Training Data: check wether the training data represents different groups.

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5. Use fairness Metrics : Metrics such as demographic parity.

6. Regular Monitoring After Deployment:

Bias should not be checked only before lanuch. They should be continuouly monitored because real-world data and user behavior can change over time.

: 7min